WW – NER Model Evaluation

Leonardo Antero SP. Yoro

IT Elective IV, BSIT

Jose Rizal University

Mandaluyong, Philippines

leonardoantero.yoro@my.jru.edu

***Abstract* — Named Entity Recognition (NER) is an important Natural Language Processing (NLP) task that aims to identify and classify entities from unstructured text. Some of these entities can include personal names, organizations, locations, and dates. In this study, we want to examine the performance of spaCy's small and medium pre-trained NER models in the sports journalism domain based upon a curated dataset of sports news articles, manually annotated to act as a gold standard for comparison. NER models made predictions based upon the manual annotations and performance was quantified using model precision, recall, F1 score, and approximate accuracy. The findings suggested that both models produced similar performance and, perhaps unsurprisingly, had success with common entities such as player names, teams, venues, and dates. There were weaknesses in predictions when confronting repeated entities, domain-specific definitions, and complex multiword expressions. Additionally, the medium model offered no advantages over the small model suggesting that domain issues, and dataset specifics, may have a strong influence on NER performance. These findings identify the challenges of domain adaptation, and the influence of annotations on a specialized application of NER.**

Keywords— Named Entity Recognition, spaCy library, sports journalism, domain adaptation, NLP

I. INTRODUCTION

Named Entity Recognition (NER) is an important form of Natural Language Processing (NLP) that focuses on the recognition and classification of personal names, organizations, locations, and dates from unstructured text using computer software or tools. State-of-the-art NER systems, the most popular NER determined by the spaCy library, are easily available and demonstrated very high levels of accuracy within many different domains within unstructured general, not domain-specific, writing. Recent literature has shown that spaCy's NER capabilities use the latest machine learning tools, with efficient speed and high accuracy [1].

Although, NER system's accuracy can also change significantly based on the specificity of the domain, where terminology, naming conventions, and context are diverging from the NER's general purposes, broader training corpus. For example, Sports Journalism typically references player names, team names, tournaments, leagues, and statistical terms that are often diverging from the spaCy NER training data corpus. This discrepancy (keenly seen in the example sports journalism) highlights the major importance of assessing spaCy's NER model within the sports domain in order to establish how adaptive and flexible the model is, while also demonstrating the challenges faced to extract domain-specific entities from NER.

This research evaluates the NER performance of spaCy's NER in the sports domain by assembling a dataset of sports news articles, and labeling the entities ourselves to create a gold standard for comparison. The predictions that spaCy produces are systematically compared with the tagging done manually, and its performance assessed based on accuracy, precision, recall, and F1-score. The systematic comparison allows us to quantitatively evaluate the performance of the model and discover where in-domain adaptation may be needed.

II. METHODOLOGY

a) Domain Description and Curation

The goal of this phase was to find a suitable application domain in which to apply Named Entity Recognition (NER), as well as to compile a representative dataset to test the models. The sports domain was chosen because of its broad range of terminologies including player names, team names, tournament names, and statistical terms, allowing for a comprehensive implementation of NER. A dataset was obtained from "News Articles Classification Dataset for NLP and ML" on Kaggle [2], specifically, sports_data.csv. In total, four separate text samples were selected from the sports category; specifically, the examples selected were from rows 1693, 932, 609, and 24. The text samples were chosen for their length and named entity diversity, which provide a meaningful basis for comparing performance in identifying and classifying the relevant named entities in sports journalism.

XXX-X-XXXX-XXXX-X/XX/$XX.00 ©20XX IEEE

SAMPLE 1

As LeBron James inches closer to scaling the summit of the NBA's all-time points scoring record, tickets prices at venues where he is likely to get there are going through the roof as well. Tickets for the Los Angeles Lakers versus Oklahoma City Thunder game on February 7 at the Crypto.com Arena are selling for an eye-watering sum of $92,000 (approximately Rs 75 lakh), according to Bleacher Report.

This, despite the fact that it's not a certainty that James will break the record in the Lakers-Thunder game. The Lakers will take on New Orleans Pelicans on February 4, the Milwaukee Bucks on February 9 and the Golden State Warriors on February 11 — each of these three games could also see James break the record.

The average price for a ticket in the Lakers-Thunder game has also gone up to $1,152 (approximately Rs 94,000) which is a whopping 390 percent increase in prices since the tickets initially went on sale, according to CBS Sports.

The average price for the Lakers-Bucks game is even more steep: $1,302 (just over Rs one lakh) which is a 175 percent jump from the start of the season.

Top Sports News Now

Breaking: KL Rahul ruled out of third Test against England, fellow Karnataka batsman to replace him

For the third time in the last seven months, Australia defeat India in an ICC final

How Xabi Alonso stopped the goal machine Harry Kane against Bayern Munich, and is making heads turn steering the wheel at Bayer Leverkusen

CLICK HERE FOR MORE

Despite having a sore left ankle before the game started, James scored 26 points against the Indiana Pacers — which was his 34th consecutive game with at least 20 points — late on Thursday to stand just 63 points away from breaking Kareem Abdul-Jabbar's record of 38,387 points. Abdul-Jabbar has held the record since before the 38-year-old James was born. Barring his first season as a rookie for the Cleveland Cavaliers in 2003-04, the 19-time All-Star has racked up a season average of at least 25 points per game in every other season of his career.

Fig. 1. Row 1693 Article

SAMPLE 2

Seven-time Formula One champion Lewis Hamilton insisted he will not be silenced by offseason rules that prohibit drivers from speaking out on political issues.

Hamilton said Wednesday he will continue to use his global platform to promote his wide-ranging interests, which include social justice and race, human rights and protection of the LBGTQ community.

The FIA in December updated its International Sporting Code to require prior written permission for drivers to make or display "political, religious and personal statements or comments" during race weekends. The FIA is the governing body for Formula One.

Mercedes' Lewis Hamilton during a press conference. (Reuters)

The crackdown on free speech has been condemned by most drivers but Hamilton only weighed in for the first time Wednesday, when Mercedes revealed its 2023 car.

Hamilton said the FIA crackdown "doesn't surprise me" but he quickly dismissed it when he learned about it over the offseason.

"Nothing will stop me from speaking on the things that I feel that I'm passionate about and issues there are," Hamilton said. "I feel the sport does have a responsibility, still, always, to speak out as a means to create awareness on important topics, particularly as we are traveling to all these different places, so nothing changes."

Asked if he was prepared to be penalized by the FIA for violating the new rule, Hamilton said "it would be silly to say I would want to take extra penalty points" but remained steadfast in that he won't be silenced.

ADVERTISEMENT

"I'm still going to be speaking on my end," Hamilton said. "We still have this platform. There's a lot of things we need to tackle."

Most of the drivers have spoken out against the new rule and most recently by F1 boss Stefano Domenicali, who recently told The Guardian newspaper the series would not be imposing any sort of gag. Domenicali said he expected the FIA to soon clarify its position.

Hamilton is the most vocal driver in F1 and remains the same change agent 17 years into his career as when he became the first Black winner in F1 in 2008. The British racer is now 38 years old, the winningest driver in series history and is tied with Michael Schumacher with a record seven titles.

Hamilton remains the only Black driver at the most elite level of motorsports.

ADVERTISEMENT

Hamilton often speaks out while racing in countries with questionable human rights records, or when an issue arises in which he feels his voice can lend support, which would be banned under the new FIA rule.

Hamilton last year sparred with the FIA over its crackdown of drivers wearing jewelry in the car and mocked the rule by arriving at a news conference wearing three watches, eight rings and multiple necklaces. Hamilton and the FIA had a protracted back-and-forth over the jewelry ban in which he received an extension on a deadline to remove some piercings; the two sides eventually came to an agreement.

Top Sports News Now

India vs Australia Live Score, U19 World Cup 2024 Final

In U-19 cricketers' journey to WC final, meet those who laid ground for them

Chelsea all at sea: Ponderous Pochettino, misfiring stars despite £1 billion in transfers and clueless American owner

CLICK HERE FOR MORE

Hamilton's teammate George Russell is a director of the Grand Prix Drivers' Association, which represents F1 drivers, and shared the criticism of the FIA's attempt to limit drivers' speech.

"I think it's totally unnecessary in the sport and in the world we live at the moment. Naturally, we are obviously seeking clarification and I trust it will be resolved," Russell said Wednesday.

Fig. 2. Row 932 Article

SAMPLE 3

Ferrari's long-serving race strategy engineer Ravin Jain has taken over from Spaniard Inaki Rueda as the head of strategy of the iconic car-maker's F1 team.

Jain will begin the high-pressure job at the season-opening Bahrain Grand Prix next weekend. Reports say the Ferrari F1 team management decided to promote Jain following quite a few high-profile errors last season.

Ferrari did well last season with Charles Leclerc finishing second behind title winner Max Verstappen of Red Bull. Their other driver George Russell was fourth. Some of the questionable calls made last year included a bad pit strategy during the Monaco Grand Prix which cost Leclerc a potential win, while at the Brazilian Grand Prix, he was the only driver on intermediate tyres for the start of Qualification 3.

In the backdrop of the two questionable decisions, Ferrari's new team principal Fred Vasseur, who took over from Mattia Binotto, decided to ease out Rueda from the role and make the Oxford-graduate Indian with a first-class degree in physics, the head of strategy.

Jain joined Scuderia Ferrari in 2017 and, over the last six years, rose to the post of Race Strategy Engineer. Rueda will continue to be a part of the team with a factory-based sporting role at Ferrari's headquarters in Maranello.

🚨 | BREAKING

Introducing Ferrari's new chief strategist, Ravin Jain 😍

Of Indian origin, and brought up in England, Ravin has a Physics undergraduate degree, and a master's in Mathematical and Theoretical Physics, both from University of Oxford.

pic.twitter.com/hsEg7XK1Xz

— Ferrari News 🐎 (@FanaticsFerrari) February 23, 2023

Early this year Vasseur had indicated he would make changes in Ferrari's strategy team, saying he was in the "process of reviewing everything".

"We are in the process to review everything. It's a bit short notice for me, but we'll have to do some improvement," Vasseur had said in January this year.

Fig. 3. Row 609 Article

SAMPLE 4

On Tuesday, Shubman Gill was named India's men cricketer of 2023. On Wednesday, a day before India's first Test against England, he turned up for a full-on nets session at the Rajiv Gandhi International Stadium in Hyderabad. It's a rare sight because most first-eleven regulars skip the match-eve practice session. Mostly, only the back-ups turn up, more in the hope of an opportunity opening up at the last minute.

But few batsmen in India's line-up, weakened by the absence of Virat Kohli, would be as under pressure as Gill could be. So he slogged for a good couple of hours in the nets under the unusually searing January sun. After starting off against seamers, he quickly shifted to facing the spinners (all net bowlers). Having struggled in the past against off-spinners, he would seek one out and instruct him to bowl a bit slower through the air as he alternated between attack and defence. There were times when he was beaten on the outside edge. Gill would often respond with an aggressive shot.

Fig. 4. Row 24 Article

b) Manual Data Annotation (Ground Truth Generation)

It is important to create a "gold standard" or ground truth for evaluating Named Entity Recognition (NER) models because it allows for a definitive place to compare against model predictions. With a well-annotated dataset, researchers can review the actual performance, accuracy, precision, recall, and F1-score of the NER system. Having a gold standard allows for a consistent and standardized meaning of tagging. It is important for the gold standard to be of high quality since any inconsistency or inaccuracy will produce misleading performance metrics, and therefore misrepresent the reliability of the model. The gold standard process not only informs about the model's strengths and weaknesses but also gives guidance for future improvements and adaptations to particular domains. A tendency for the quality of the ground truth contributes to a high impact on the reliability of performance metrics faced in NER tasks, a potent argument for careful annotation practice [3].

In this study, the named entities in the curated texts were manually annotated using spaCy's built-in entity types: PERSON, ORG, LOC, DATE, GPE, and other tags to ensure consistency. Each entity was tagged precisely, with clear boundaries established for labeling, to act as a standard of truth for evaluation. The following figures illustrate samples of selected words that were manually tagged as part of the gold standard.

SampleID	EntityText	EntityLabel
1	LeBron James	PERSON
1	James	PERSON
1	KL Rahul	PERSON
1	Karnataka	GPE
1	Xabi Alonso	PERSON
1	Harry Kane	PERSON
1	Kareem Abdul-Jabbar	PERSON
1	Abdul-Jabbar	PERSON
1	England	NORP
1	Australia	NORP
1	India	NORP
1	Crypto.com Arena	FAC
1	NBA	ORG
1	the Los Angeles Lakers	ORG
1	Oklahoma City Thunder	ORG
1	Bleacher Report	ORG
1	Lakers	ORG
1	New Orleans Pelicans	ORG
1	the Milwaukee Bucks	ORG

Fig. 5. Article 1693 Manual Tag Sample

2	Lewis Hamilton	PERSON
2	Hamilton	PERSON
2	Stefano Domenicali	PERSON
2	Domenicali	PERSON
2	Michael Schumacher	PERSON
2	George Russell	PERSON
2	Russell	PERSON
2	Pochettino	PERSON
2	LBGTQ	NORP
2	Black	NORP
2	British	NORP
2	American	NORP
2	Formula One	ORG
2	FIA	ORG

Fig. 6. Article 932 Manual Tag Sample

3	Ravin Jain	PERSON
3	Rueda	PERSON
3	Charles Leclerc	PERSON
3	Max Verstappen	PERSON
3	George Russell	PERSON
3	Fred Vasseur	PERSON
3	Mattia Binotto	PERSON
3	Inaki Rueda	PERSON
3	Jain	PERSON
3	Spaniard	NORP
3	Indian	NORP
3	Ferrari	ORG
3	Ferrari F1 team	ORG

Fig. 7. Article 609 Manual Tag Sample

4	Shubman Gill	PERSON
4	Virat Kohli	PERSON
4	Gill	PERSON
4	Rajiv Gandhi International Stadium	FAC
4	India	ORG
4	England	ORG
4	Hyderabad	GPE
4	Test	EVENT
4	Tuesday	DATE
4	2023	DATE
4	January	DATE

Fig. 8. Article 24 Manual Tag Sample

c) Automated Entity Recognition with spaCy

The spaCy small (en_core_web_sm) and medium (en_core_web_md) English models were used to evaluate consistency over multiple model sizes. The implementation started with the loading of the spaCy model and modules needed for listing all of the required extra information for storing the entities, including csv, under error conditions when a model is not present. Once everything loaded, the pipeline printed the complete list of NER entity categories to assist in the reference list for possible labels.

The curated texts were processed through a programmatic user interactive loop, where the user would input each text, confirm it was not blank, and analyze it. For each text sampled, the spaCy pipeline identified named entities, and printed the entity text, label, and definition. Each identified entity was simultaneously stored in a list along with a sample counter in order to keep everything in order for later comparison.

```python
import spacy
from spacy import displacy
import os
import datetime
import csv

# Install spacy and download a language model (if you haven't already) --
# (en_core_web_sm is a small English model, suitable for demonstration)

try:
    nlp = spacy.load("en_core_web_sm")
    print("spaCy model 'en_core_web_sm' loaded successfully.")
except OSError:
    print("Error: 'en_core_web_sm' model not found. Please run 'python -m spacy download en_core_web_sm' in your terminal.")
    exit()

# --- Main execution block ---
if __name__ == "__main__":
    print("\n--- Named Entity Recognition with spacy ---")

    # --- Display available NER entity categories ---
    if 'ner' in nlp.pipe_names:
        ner_labels = nlp.get_pipe('ner').labels
        print("\n--- Available Entity Categories (Labels) ---")
        print(", ".join(sorted(ner_labels)))
    else:
        print("\NNER pipeline component not found in the loaded model.")

    print("\nEnter the text you want to analyze (type 'quit' to exit):")
    output_counter = 1
    all_entities = []  # List to store all detected entities

    while True:
        input_text = input("\nEnter text to process (enter quit to stop): ")

        if input_text.lower() == 'quit':
            break
        elif not input_text.strip():
            print("Invalid input: Please provide a non-empty string for processing.")
            continue

        doc = nlp(input_text)

        print("\n--- Detected Named Entities ---")
        if doc.ents:
            for ent in doc.ents:
                print(f"Text: '{ent.text}' | Label: {ent.label_} ({spacy.explain(ent.label_)})")
                print()
                all_entities.append((output_counter, ent.text, ent.label_))  # Append to the list
            output_counter += 1  # Increment the counter after processing
```

Fig. 9. Detected Named Entities Display, Samples Loop and Loop Counter

```python
    if output_counter % 5 == 0:
        # Save to CSV
        csv_filename = "spacy_predictions.csv"
        with open(csv_filename, 'w', newline='', encoding='utf-8') as csvfile:
            writer = csv.writer(csvfile)
            writer.writerow(['SampleID', 'EntityText', 'EntityLabel'])
            writer.writerows(all_entities)

        print(f"\nAll detected entities have been saved to '{csv_filename}'.")
        all_entities.clear()

else:
    print("No named entities detected in the text.")
```

Fig. 10. Predictions to CSV file Conversion

```
spaCy model 'en_core_web_sm' loaded successfully.

--- Named Entity Recognition with spacy ---

--- Available Entity Categories (Labels): ---
CARDINAL, DATE, EVENT, FAC, GPE, LANGUAGE, LAW, LOC, MONEY, NORP, ORDINAL, ORG, PERCENT, PERSON, PRODUCT, QUANTITY, TIME, WORK_OF_ART

Enter the text you want to analyze (type 'quit' to exit):

Enter text to process (enter quit to stop): As LeBron James inches closer to scaling the summit of the NBA's all-time points scoring record, tickets

--- Detected Named Entities ---
Text: 'LeBron James' | Label: PERSON (People, including fictional)

Text: 'NBA' | Label: ORG (Companies, agencies, institutions, etc.)

Text: 'the Los Angeles Lakers' | Label: ORG (Companies, agencies, institutions, etc.)

Text: 'Oklahoma City Thunder' | Label: GPE (Countries, cities, states)

Text: 'February 7' | Label: DATE (Absolute or relative dates or periods)

Text: 'the Crypto.com Arena' | Label: FAC (Buildings, airports, highways, bridges, etc.)

Text: '92,000' | Label: MONEY (Monetary values, including unit)

Text: 'Bleacher Report' | Label: ORG (Companies, agencies, institutions, etc.)

Text: 'James' | Label: PERSON (People, including fictional)

Text: 'New Orleans' | Label: NORP (Nationalities or religious or political groups)

Text: 'February 4' | Label: DATE (Absolute or relative dates or periods)

Text: 'the Milwaukee Bucks' | Label: EVENT (Named hurricanes, battles, wars, sports events, etc.)

Text: 'February 9' | Label: DATE (Absolute or relative dates or periods)

Text: 'the Golden State Warriors' | Label: FAC (Buildings, airports, highways, bridges, etc.)

Text: 'February 11' | Label: DATE (Absolute or relative dates or periods)
```

Fig. 11. spaCy Predictions Output from Inputted Sample

The program had a routine for dealing data, every time there were four processes last output for the entities, it automatically dumped the entities in a CSV file from each run. Each CSV file consisted of the columns SampleID, EntityText, and EntityLabel, thus, the output is formatted and arranged according to the gold standard evaluation criteria.

spacy_predictions.csv

1 to 50 of 188 entries Filter

SampleID	EntityText	EntityLabel
1	LeBron James	PERSON
1	NBA	ORG
1	the Los Angeles Lakers	ORG
1	Oklahoma City Thunder	GPE
1	February 7	DATE
1	the Crypto.com Arena	FAC
1	92,000	MONEY
1	Bleacher Report	ORG
1	James	PERSON
1	New Orleans	NORP
1	February 4	DATE
1	the Milwaukee Bucks	EVENT
1	February 9	DATE
1	the Golden State Warriors	FAC
1	February 11	DATE
1	three	CARDINAL
1	James	PERSON
1	up to $1,152	MONEY
1	94,000	CARDINAL
1	390 percent	PERCENT
1	CBS Sports	ORG
1	the Lakers-Bucks	ORG
1	1,302	MONEY

Fig. 12. CSV File Content

This method, meanwhile, provided a reproducible and consistent way to extract named entities and create our dataset of spaCy predictions that could be directly compared to manually annotated entities to derive performance metrics like accuracy, precision, recall, and F1-score. Also, utilizing both small and medium models enabled comparisons of model behavior based on differences between training size and vocabulary coverage, specifically to sports, for both small and medium models.

d) Performance Analysis and Accuracy Assessment

The accuracy of the spaCy NER model was assessed quantitatively by comparing the automatic delineation of entities made by the model with the gold standard that was hand labelled. This required two separate scripts that were written in Python. The first script combined the hand labelled dataset with the CSV of spaCy predictions on SampleID and EntityText to align entity occurrences and identify errors where entities might have been excluded or annotated differently. This combine was done with the output of each of the spaCy models, the small (en_core_web_sm) model and the medium (en_core_web_md) model, to see if there were differences between the performance of the models that impacted their accuracy.

```python
import pandas as pd

manual_labels_path = 'YORO_NER.csv'
spacy_preds_path = 'spacy_predictions.csv'

manual_df = pd.read_csv(manual_labels_path, encoding='ISO-8859-1')

manual_df = manual_df[['SampleID', 'EntityText', 'EntityLabel']]

spacy_df = pd.read_csv(spacy_preds_path, encoding='ISO-8859-1')


# Merge on SampleID and EntityText to align entity occurrences
merged = pd.merge(
    manual_df,
    spacy_df,
    on=['SampleID', 'EntityText'],
    how='outer',
    suffixes=('_manual', '_spacy')
)

# Fill NaN labels with explicit string for clarity
merged['EntityLabel_manual'] = merged['EntityLabel_manual'].fillna('NOT LABELED')
merged['EntityLabel_spacy'] = merged['EntityLabel_spacy'].fillna('NOT LABELED')

# Identify mismatches: where labels differ
mismatches = merged[merged['EntityLabel_manual'] != merged['EntityLabel_spacy']]

# Summary stats
total_manual = len(manual_df)
total_spacy = len(spacy_df)
total_matches = len(merged) - len(mismatches)
total_mismatches = len(mismatches)

print(f"Total manual entities: {total_manual}")
print(f"Total spaCy predicted entities: {total_spacy}")
print(f"Total matched entities (including exact label matches and missing): {len(merged)}")
print(f"Total mismatches (label differences or missing): {total_mismatches}")

print("\nSample mismatches:")
print(mismatches.head())
```

Fig. 13. First Code for Comparison

The second script calculated the standard NER metrics of precision, recall, F1-score and approximated accuracy. The true positives were considered to be all of the entities which matched SampleID, EntityText and EntityLabel from the gold standard and the spaCy predictions. The false positives were those predicted by spaCy but either missing or different in the manual annotations, while false negatives were defined as being missing in the spaCy predictions but existing in the gold standard. These counts were used in calculations for precision, recall and F1-score, with accuracy approximated as the proportion of correctly classified entities to total entities.

```python
from sklearn.metrics import precision_score, recall_score, f1_score

# For metrics, we consider only entities that appear in either manual or spaCy sets.
# Let's define:
# - True Positives (TP): entities with matching SampleID, EntityText, AND matching EntityLabel
# - False Positives (FP): entities predicted by spaCy but not in manual OR label mismatch
# - False Negatives (FN): entities in manual but missing or mislabeled in spaCy

merged['exact_match'] = merged['EntityLabel_manual'] == merged['EntityLabel_spacy']

# Count True Positives: exact matches where spaCy predicted and manual has entity
TP = merged[(merged['EntityLabel_manual'] != 'NOT LABELED') &
            (merged['EntityLabel_spacy'] != 'NOT LABELED') &
            (merged['exact_match'])].shape[0]

# False Positives: spaCy predicted something that manual didn't label or mismatched label
FP = merged[(merged['EntityLabel_manual'] == 'NOT LABELED') |
            ((merged['EntityLabel_manual'] != merged['EntityLabel_spacy']) &
             (merged['EntityLabel_spacy'] != 'NOT LABELED'))].shape[0]

# False Negatives: manual labeled but spaCy missed or mislabeled
FN = merged[(merged['EntityLabel_spacy'] == 'NOT LABELED') |
            ((merged['EntityLabel_manual'] != merged['EntityLabel_spacy']) &
             (merged['EntityLabel_manual'] != 'NOT LABELED'))].shape[0]

print(f"True Positives (TP): {TP}")
print(f"False Positives (FP): {FP}")
print(f"False Negatives (FN): {FN}")

# Precision = TP / (TP + FP)
precision = TP / (TP + FP) if (TP + FP) > 0 else 0

# Recall = TP / (TP + FN)
recall = TP / (TP + FN) if (TP + FN) > 0 else 0

# F1-Score = 2 * (precision * recall) / (precision + recall)
f1 = 2 * precision * recall / (precision + recall) if (precision + recall) > 0 else 0

print(f"\nPrecision: {precision:.4f}")
print(f"Recall: {recall:.4f}")
print(f"F1-Score: {f1:.4f}")

total_entities = merged.shape[0]
correct_entities = TP
accuracy = correct_entities / total_entities if total_entities > 0 else 0
print(f"Approximate Accuracy: {accuracy:.4f}")
```

Fig. 14. Second Code for Comparison

Use of these metrics allows for a standardized assessment of the performance of NER models, yielding information about the correctness and completeness of model-generated predictions. Although we present a more complete discussion of results in the following section, these calculations are instrumental in identifying the strengths and weaknesses of spaCy's pre-trained models when applied to a specific domain like sports news and for guiding next steps in improving and adapting models for the domain-specific NER applications in the future [4].

III. Results

The performance of the spaCy NER models quantitatively assessed against the handcrafted gold standard for the sports news dataset. Table I provides an overview of the quantitative metrics found in the results such as true positives, false positives, false negatives, precision, recall, F1-score, and estimated accuracy for both models--the small (en_core_web_sm), and the medium (en_core_web_md). There were also sample discrepancies observed between the predicted labels and the manual labels that displayed the types of entities that were misidentified, or misclassified, and also where identified entities were missed altogether.

Metric	sm	md
Total Manual Entities	140	140
Total SpaCy Predicted	188	188
Total Matches	225	225
Total Mismatches	109	109
True Positives (TP)	116	116
False Positives (FP)	72	72
False Negatives (FN)	65	65
Precision	0.6170	0.6170
Recall	0.6409	0.6409
F1-Score	0.6287	0.6287
Approx. Accuracy	0.5156	0.5156

Table 1. spaCy sm and md model performance comparison to gold standard

```
Sample mismatches:
   SampleID EntityText EntityLabel_manual EntityLabel_spacy
0         1      1,302        NOT LABELED             MONEY
1         1       1152              MONEY       NOT LABELED
2         1       1302              MONEY       NOT LABELED
5         1         20           CARDINAL       NOT LABELED
7         1         25           CARDINAL       NOT LABELED
```

Fig. 15. Sample Mismatches for spaCy small model

```
Sample mismatches:
   SampleID EntityText EntityLabel_manual EntityLabel_spacy
0         1      1,302        NOT LABELED             MONEY
1         1       1152              MONEY       NOT LABELED
2         1       1302              MONEY       NOT LABELED
5         1         20           CARDINAL       NOT LABELED
7         1         25           CARDINAL       NOT LABELED
```

Fig. 16. Samples Mismatches for spaCy medium model

An examination of the sample mismatches indicates that numeric values such as “1,302”, “1152”, and “1302” were mislabelled and inconsistently tagged, frequently mislabelled as MONEY or CARDINAL by spaCy, whereas the gold standard had primarily tagged the values correctly or did not tag them at all. The models gave a moderate precision and recall approximately 0.52 accuracy, meaning about half the entities were established. In entirety, the summary provides a benchmark for assessing the spaCy model performance on sports text in the specific domain and will be discussed later in the Discussion section.

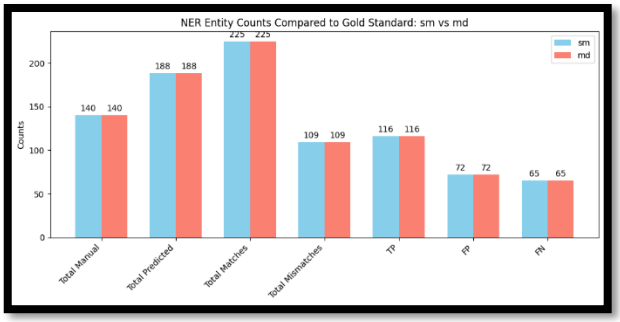

Fig. 17. NER Entity Counts of both models compared to Gold Standard

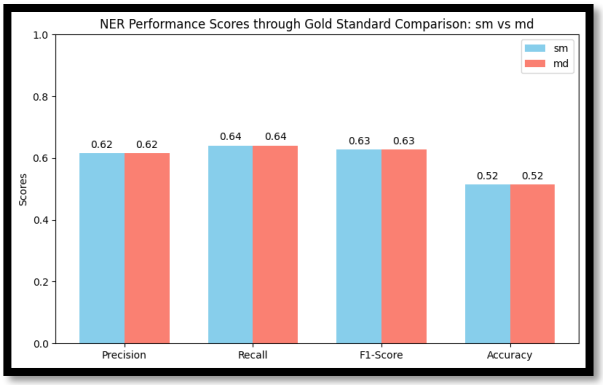

Fig. 18. NER Performance Scores of both models through Gold Standard Comparison

IV. Discussion

a) Overview of Model Performance

Conducted experimentations into spaCy's standard (en_core_web_sm) and medium (en_core_web_md) NER models showed almost identical results and had an overall agreement of approximately 0.52. Precision (0.617) and recall (0.641) both had an F1-score of 0.629 on both models. The medium model had a size advantage and had learned richer representations of word vectors, but the models produced almost the same extraction. This likely occurred because of the small sample size and discipline-specific documents that consisted of mainly an athlete's name, sports team name, venue, date, and numeric values. Since these entities are well captured with spaCy pretrained knowledge and the standard model, the medium NER model did not make any differences in named entity recognition.

b) Strengths of the Models

Both models achieved consistent performance in correctly predicting more easily recognizable entity types including athlete and coach names also like "LeBron James" and "Lewis Hamilton", organizations and teams like "NBA" and "Mercedes", venues like "Crypto.com Arena" and date or events like "4-Feb" and "Bahrain Grand Prix". The results suggest the pretrained models were adept for general widely known entities even though they were still within a specific domain of sports news.

c) Weaknesses and Sources of Error

A few error patterns were found. spaCy tagged repeated entities in the text multiple times while those repeated entities were labeled only once in the manual gold standard. As a result, the discrepancies inflated the false positives and complicated the accuracy measurements because the entity "James" would come again and again in the span of a single prediction. In addition, domain or less talked about entities, including multiword regulated expressions, and tournament reference entity types, were sometimes missed or misclassified altogether. Moreover, some entities were tagged as unexpected types, for example, Formula was tagged as PERSON. In part this is because of approaches that drove contextual ambiguity in the case of nouns or because they general-purpose pretrained models often struggle at capturing sports nouns, or unexpected types altogether.

d) Implications for Domain-Specific NER

These findings raise important implications for using general domain NER models in specialized domains. Merely increasing model size will not improve performance if the dataset is small and mostly contains common knowledge entities. The coverage of the pretrained model is also an important consideration, and specialized domains with specific language, such as sporting language and terminologies, may need a combination of pretraining and fine-tuning the NER model, or training a custom NER model altogether to improve performance. Additionally, if there are differences in how repeated entities are counted it must be considered for metric interpretation, as these differences will heavily affect TP, FP, FN and downstream metrics like precision, recall, and F1-score. Identifying these limitations provide useful information for researchers to plan future domain adaptation approaches, such as increasing the amount of training data, including custom labels, or performing further entity normalization.

e) Justification of Low Accuracy

Although the approximate accuracy of 0.5156, the same for both the small and medium spaCy models, may seem low, there are some reasons specific to this study that rationalize these results.

Firstly, the gold standard tags are all independently tagged entities, and manual coding does not capture repeated instances of the same entity. In contrast, spaCy replicates every observation of an entity that it encounters. This difference raises false positives while lessening the apparent accuracy even if the model is recognizing repeated entities correctly.

Secondly, the dataset includes a small number of sports news articles with sports-specific terms, such as team abbreviations, tournament titles, and multiword player mentions that were not well-represented in spaCy's pretrained models. For this reason, some entities that were tagged were misclassified, omitted or simply labelled differently to what the model produced, leading to different metrics.

Lastly, as the medium model generated no additional correct predictions to the small model, the low accuracy simply reflected the lack of applicability of heuristic NER models packaged for general purposes in specialized domains of knowledge without finetuning. This finding highlights the need to consider domain specificity, dataset size, and handling repeated entities as we interpret NER performance metrics [4].

f) Handling of spaCy Predictions

For consistency and reproducibility of analyses, the gold standard manually annotated data and the spaCy model predictions were both saved and processed in .csv format. While converting to .xlsx was an option, this would only complicate matters for no effect on evaluating model performance. Having both datasets in CSV format ensured that they could be merged properly, and mitigated the chance for formatting errors, and letting me focus on a common goal: quantitatively comparing spaCy predictions relative to the gold standard to yield precision, recall, F1-score, and approximate accuracy.

V. CONCLUSION

This research tested the performance of spaCy's small (en_core_web_sm) and medium (en_core_web_md) pre-trained NER models on a manually-coded sports news dataset [1], [2]. A gold standard was constructed to provide a reliable benchmark [3], and standard NER metrics for precision, recall, F1_score, and accuracy were measured.

Both models showed a similar approximate accuracy of 0.52, with similar metrics for precision and recall; both demonstrated strengths in recognizing more publicly prominent entities such as player names, teams, locations, and dates. Both models also had weaknesses in recognizing repeated entities, domain-specific language, and more difficult multi-word expressions, which also introduced false positives, false negatives, and label mismatches [4].

Crucially, the medium model even produced no superior performance than the small model, underlining that reliance only on model size is not a guarantee for performance with domain-specific tasks. These results further emphasized the value of domain adaptation, annotation process, and metrics when using models created for broad tasks and applied to specialist text corpora [3, 4].

Future work could investigate the fine-tuning of a pre-trained model on a specific sport corpus or an ensemble approach to enhance recognition accuracy and in turn reduce mismatches in specialists' domains.

VI. REFERENCES

[1] "NLP Application: Named Entity Recognition (NER) in Python with SpaCy," Analytics Vidhya, June 2021. [Online]. Available: https://www.analyticsvidhya.com/blog/2021/06/nlp-application-named-entity-recognition-ner-in-python-with-spacy. [Accessed: Aug. 2025].

[2] Banuprakash V., "News Articles Classification Dataset for NLP and ML," Kaggle, [Online]. Available: https://www.kaggle.com/datasets/banuprakashv/news-articles-classification-dataset-for-nlp-and-ml?select=sports_data.csv. [Accessed: Aug. 2025].

[3] Gurevych, I., et al., "The importance of ground truth in the evaluation of named entity recognition," Biodiversity Data Journal, vol. 9, 2021. [Online]. Available: https://bdj.pensoft.net/article/89481/instance/7788834. [Accessed: Aug. 14, 2025].

[4] Encord, "Named Entity Recognition: What it is and why it matters," Encord Blog. [Online]. Available: https://encord.com/blog/named-entity-recognition . [Accessed: Aug. 14, 2025].